When Social Choice Creates Social Polarization

IE 305 Simulation Lab Project

Mihir Vasave (24B4517) Vaibhav Mathur (24B4534)

1. Problem & System Overview

People do not keep the same social connections forever. We tend to interact more with people we agree with, while repeated disagreement can make us less likely to interact again. At the same time, the people around us influence our opinions. This creates a feedback loop that can turn a mixed social network into separate communities.

Our question

Can a group that starts out socially mixed become polarized simply because people change who they interact with?

How the system works

- Each person has an opinion on a fictional controversial topic.
- People interact through a social network.
- An interaction can slightly change a person's opinion.
- Large disagreements can make a connection weaker or cause it to disappear.
- People are more likely to form new connections with others whose opinions are closer to their own.
- The changing network changes who people are exposed to, which feeds back into their opinions.

The key idea

We are not only simulating how opinions spread. We are studying the two way relationship between **what people believe** and **who they choose to interact with**.

Data Sources & Collection Plan

The model will use research based behavioural assumptions rather than personal social media data.

2. Data Sources & Collection Plan

The project is primarily a simulation study. We will use published research on opinion dynamics, social networks, homophily and adaptive networks to define the mechanisms in the model.

Inputs to the simulation

- **Initial social network:** a generated network, with alternative network structures tested for comparison.
- **Initial opinions:** a mixed population with opinions distributed across the range from one side of the issue to the other.
- **Opinion tolerance:** how much disagreement a person is willing to accept during an interaction.
- **Opinion influence:** how strongly a person's opinion responds to another person's opinion.
- **Rewiring behaviour:** the probability that a person weakens, removes or replaces a connection after disagreement.
- **Individual differences:** people can have different tolerance, activity and susceptibility to influence.

Sources

The literature base will include recent work on the interaction between information spreading, opinion polarization and network rewiring, together with established models of adaptive social networks.

Source	How it informs the model
PNAS Nexus, 2025	Information spreading, opinion change and adaptive rewiring
European Economic Review, 2025	Opinion updating with changing social connections
Chaos, Solitons & Fractals, 2024	Social rewiring and polarized communities
Physica A, 2020	Network fragmentation and societal bubbles
AnyLogic documentation	Agent connections and dynamic networks

Simulation Questions & Outputs

The goal is to identify when a mixed society stays connected and when it breaks into polarized groups.

3. Simulation Questions & Outputs

Main simulation questions

- How does the tendency to avoid people with different opinions affect polarization?
- Is there a point at which small increases in social rewiring cause the network to fragment?
- How do different initial network structures affect the final outcome?
- How does individual variation in tolerance and susceptibility affect the emergence of polarized communities?
- Under what conditions does the system remain connected and relatively mixed?

Scenarios

We will vary opinion tolerance, rewiring tendency, initial network structure and individual behavioural variation. Each configuration will be simulated repeatedly with different random seeds.

Outputs and KPIs

- **Opinion:** average opinion, opinion spread and degree of polarization.
- **Network:** average number of connections, clustering and number of connected communities.
- **Interaction:** frequency of cross-opinion interactions and number of connections removed or formed.
- **Dynamics:** time taken for polarization or fragmentation to emerge.
- **Monte Carlo:** probability of polarization and fragmentation across repeated runs.

Expected insight

The simulation will show how individual decisions about interaction can create a collective outcome that nobody explicitly intended: a connected population can gradually separate into communities with increasingly different opinions.

Implemented in AnyLogic using agent based modelling, stochastic interactions, dynamic connections and Monte Carlo experiments.